

Final Projects

- **Report (15% of grade - GROUP)**
 - Can copy + paste from proposal & checkpoints & can update/edit/change
 - Adding on:
 - Choice if we have your permission to make public
 - Overview
 - Analysis (w/ “strong” data visualizations)
 - Conclusion & Discussion
 - It’s a report, not a vomiting of data.
 - Write in the diamond structure of technical reports
 - Do not have to include deadends, massive codebases, or irrelevant explorations. Can put those in a separate file that is referenced in report.
 - Do edit for correctness & conciseness at the end
- **Video (3% of grade - GROUP)**
 - 3-5 minutes
 - Post video so that it is PUBLICLY VISIBLE on Google Drive, Dropbox, TikTok, YouTube, whatever.
 - Place a link to the video right after the title of your final project notebook
 - Everyone should participate; not everyone has to be IN the video
- **Team Evaluation Survey (1% of grade - INDIVIDUAL)**
 - Give us more information about who contributed / did not contribute; this will effect individual grades
 - Will include questions that help us with the pedagogical experiment of group dynamics running this year!

Week 10 / Finals Week Extra Credit Opportunities

- Finish strong with your group progress surveys!
 - 0.5% EC if you fill out all 7
- Fill out the teaching evaluations (SET used to be CAPE) [Evaluations site](#)
 - Must complete evaluations BEFORE 8AM on Saturday after Week 10. No exceptions.
 - 0.5% EC if >75% of the class fills this out (only happens half the time)
- Post course survey
 - 0.25% EC

Reminders

Upcoming due dates

Wed Mar 04 EDA checkpoint

Fri Mar 06 D8

How to be wrong III

Data Science in Practice

Jason G. Fleischer, PhD

Dept. of Cognitive Science

UC San Diego

Shoemaker Racing: Race or No?

Jeff Shoemaker was not sure, but his sister and partner, Amy Shoemaker, was on the phone and needed a decision. Should they run in the race or not? It had been a successful season so far, but the Sonoma race was important because of the prize money and TV exposure it promised. This first year had been hard because the team was trying to make a name for itself, and so had to run in a lot of small races. A successful outing could mean more sponsors, a chance to start making some profit for a change, and the luxury of racing only the major events. But if they suffered another engine failure on live TV...

“These engine failures are incredibly annoying,” thought Jeff. The team’s car had failed 7 times in 24 outings this season with various degrees of damage to the engine and car. No one could figure out why. It took a lot of sponsor money to replace a \$100,000 racing engine and the wasted entry fees were no small matter either. Jeff and Amy had everything they owned invested into Shoemaker Racing. This season had to be a success.

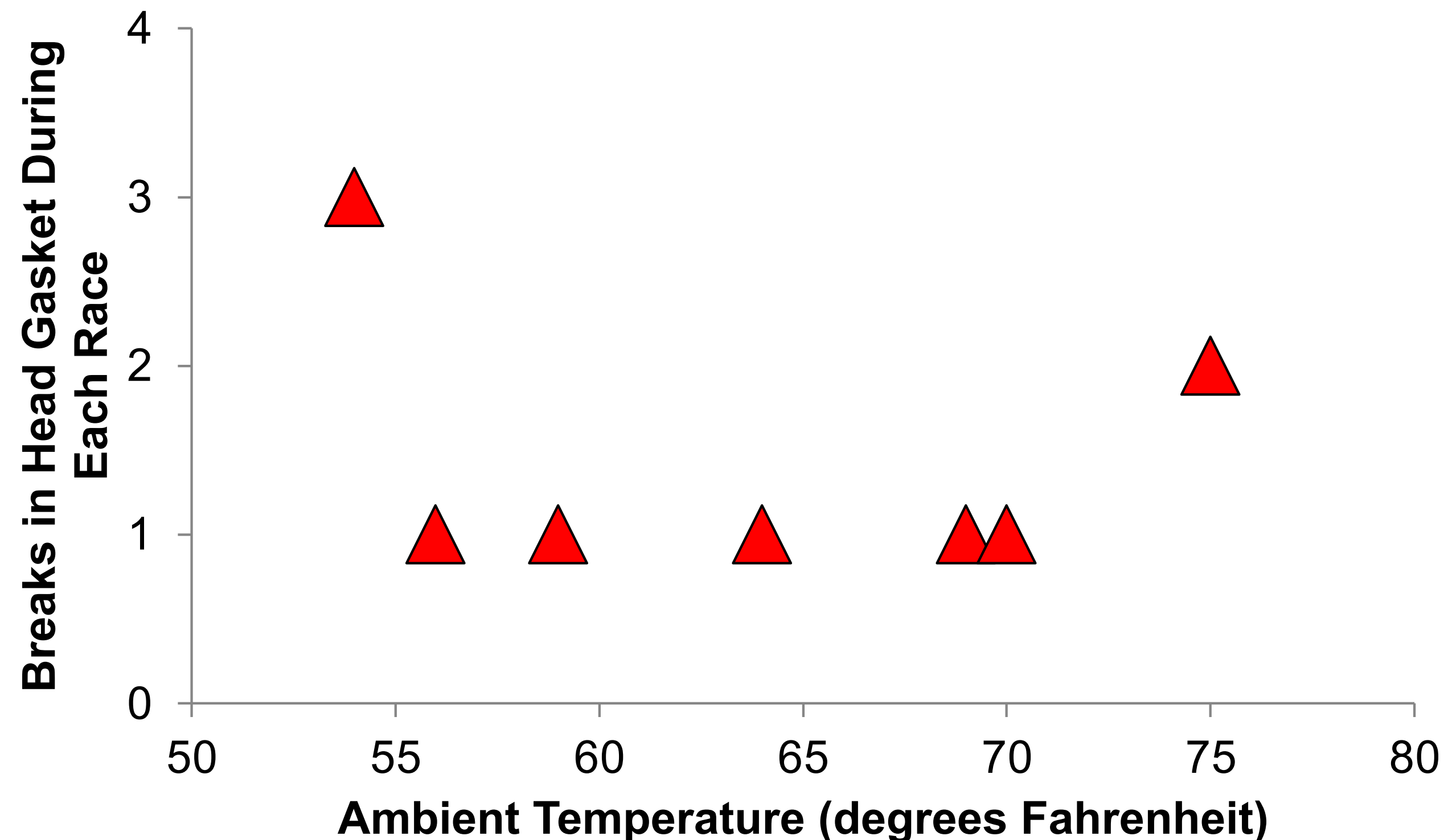
Antonio Garcia, the engine mechanic, was guessing the engine problem was related to ambient air temperature. He argued that when it was cold, the different expansion rates for the head and block were damaging the head gasket^[1] and causing the engine failures. It was below freezing last night, unusual for this time of year, which meant a cold morning for starting the race.

Tom Davis, the chief mechanic, did not agree with Antonio's "gut feeling" and had data to support his position (see Exhibit 1). He pointed out the 10 gasket failures had occurred over the entire temperature range, which meant temperature was not the issue. Tom had been racing for twenty years and believed that luck was an important element in success. "In racing, you are pushing the limits of what is known," he often said, "and that means some things are not going to be under full control. If you want to win, you have to take risks, and everybody in racing knows it. The drivers have their lives on the line, I have a career that hangs on every race, and you guys have got every cent tied up in the business. That's the thrill of it: beating the odds and winning." Last night over dinner he had added to this argument forcefully with what he called Piquet's First Law of Racing: "Nobody ever won a race sitting in the garage."

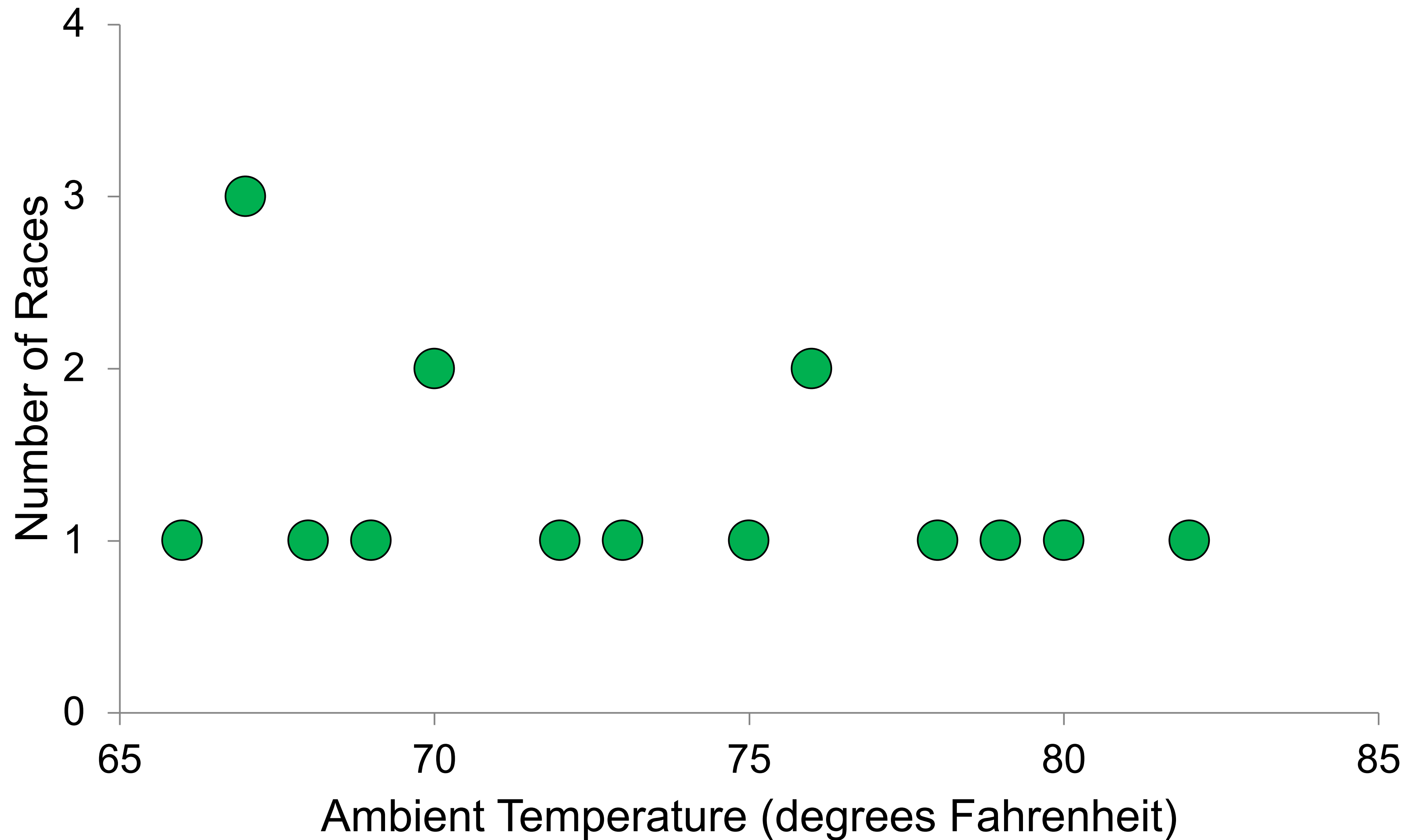
^[1] A head gasket is a mechanical seal that sits between the engine block and cylinder head(s) in an internal combustion engine. Its purpose is to seal the cylinders to ensure maximum compression and avoid leakage of coolant or engine oil into the cylinders; as such, it is the most critical sealing application in any engine, and, as part of the combustion chamber, it shares the same strength requirements as other combustion chamber components.

Note from Tom Davis

I got the data on the gasket failures from Antonio. We have run **24 races** this season, with **temperatures at race time ranging from 53 to 82 degrees**. Antonio had a good idea in suggesting we look into this, but as you can see, this is not our problem. I tested the data for a correlation between temperatures and gasket failures, and found no relationship.



Temperature and *no* gasket failure



Jeff, Amy, and Tom had discussed Shoemaker Racing's situation the previous evening. This first season was a success from a racing standpoint, with the team's car finishing in the top five in twelve of the fifteen races it completed. As a result, the sponsorship offers critical to the team's business success were starting to come in. A big break had come two weeks ago after the Auto Club race, where the team scored its fourth first-place finish. Michelin tires had finally decided Shoemaker Racing deserved its sponsorship at the Sonoma race—worth a much needed \$200,000—and was considering a full season contract for next year if the team's car finished in the top five in this race. The Michelin sponsorship was for \$5 million a year, plus incentives. Jeff and Amy had gotten a favorable response from Michelin's racing program director last week when they presented their plans for next season, but it was clear that his support depended on the visibility they generated in this race.

“Jeff, we only have another hour to decide,” Amy said over the phone. “At the end of the Auto Club race, and before we got the \$200,000 from Michelin, we were \$287,500 in the hole. If we withdraw now, we can get back half the \$75,000 entry fee for the race. We will lose Michelin, they’ll want \$125,000 of their money back, and we’ll end the season \$250,000 in the hole. If we run and finish in the top five, we have Michelin in our pocket and can add another car next season. You know as well as I do, however, that if we run and lose another engine, we are back at square one next season. We will lose the Michelin sponsorship, and a blown engine is going to lose us our current oil contract for sure. No oil company wants a national TV audience to see a smoking car being dragged off the track with their name plastered all over it. The oil sponsorship is \$2.5 million that we cannot live without. Think about it. Call Antonio and Tom if you want. But I need a decision within an hour.”

Jeff hung up the phone and looked out of the window at the crisp fall sky. The temperature sign across the street flashed “39°F, 9:23 AM.”

Jeff picked up the phone again and dialed. “Get Antonio Garcia for me.” Jeff was calling to get his engine mechanic’s opinion on whether they should run today or not. The data Tom put together indicated that temperature was not the problem, but Jeff wanted to get Antonio’s direct assessment.

Shoemaker Racing's present finances

– \$287,500 at end of Auto Club race

+ \$200,000 Michelin sponsorship

– \$75,000 Sonoma entry fee

– \$162,500



Payoffs for racing

Finish in the money

+ \$5,000,000 Michelin contract

+ \$5,000,000

Do not finish (blown engine)

– \$2,500,000 oil sponsorship

– \$100,000 replacing engine

– \$2,600,000

Finish out of the money (< 5th)

+ \$0

Withdraw

+ \$37,500	entry-fee reimbursement
– \$125,000	repay Michelin
– \$87,500	



Case facts

Racing Record

- 24 races entered
- 15 finished; 9 not finished

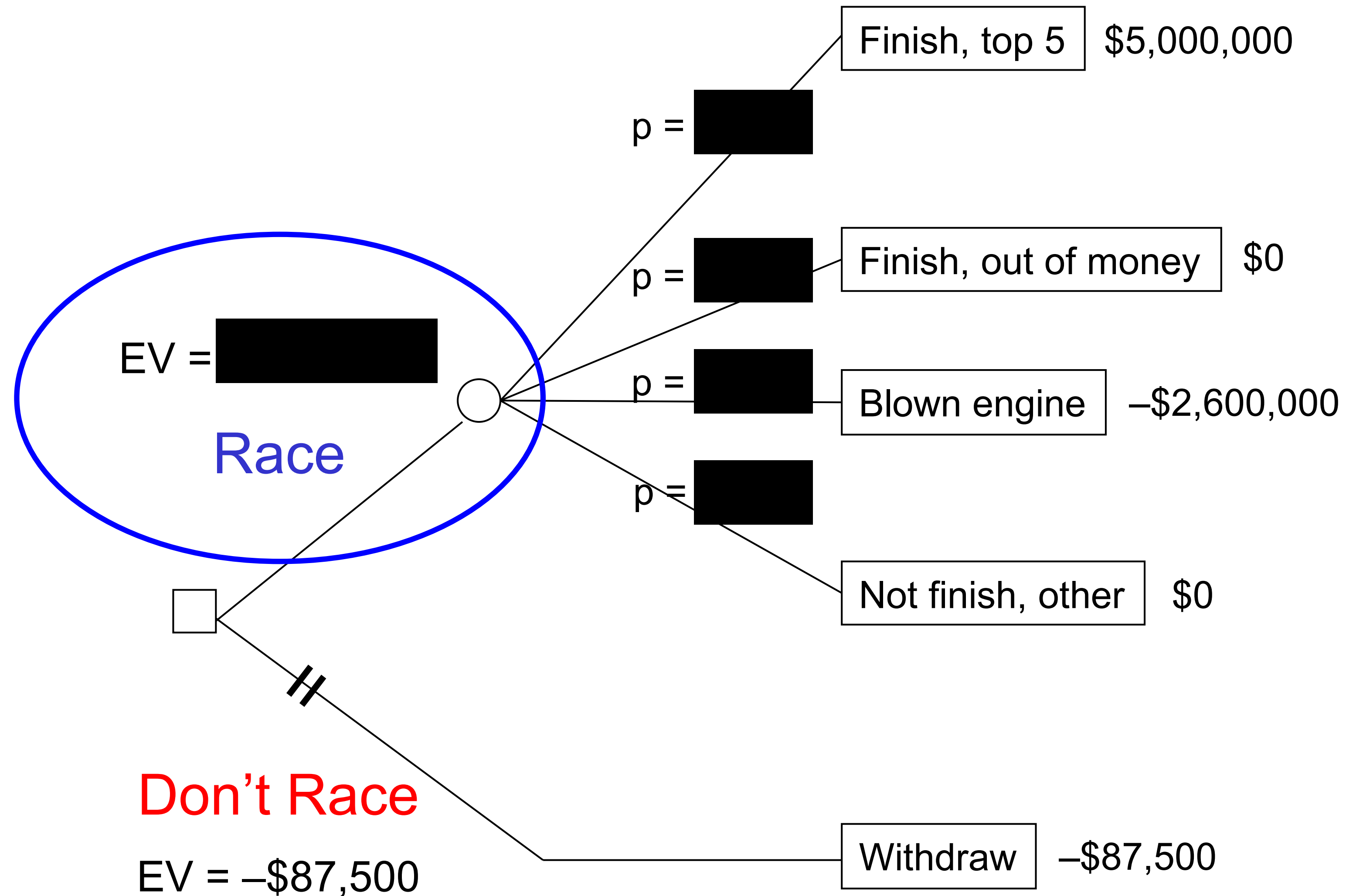
Races Finished

- 12 in Top 5 (50%)
- 3 not in Top 5 (12.5%)

Races Not Finished

- 7 with blown engine (29.2%)
- 2 for other reasons (8.3%)

Shoemaker Racing decision tree



<https://forms.gle/gLfaQVna2x81EREb9>



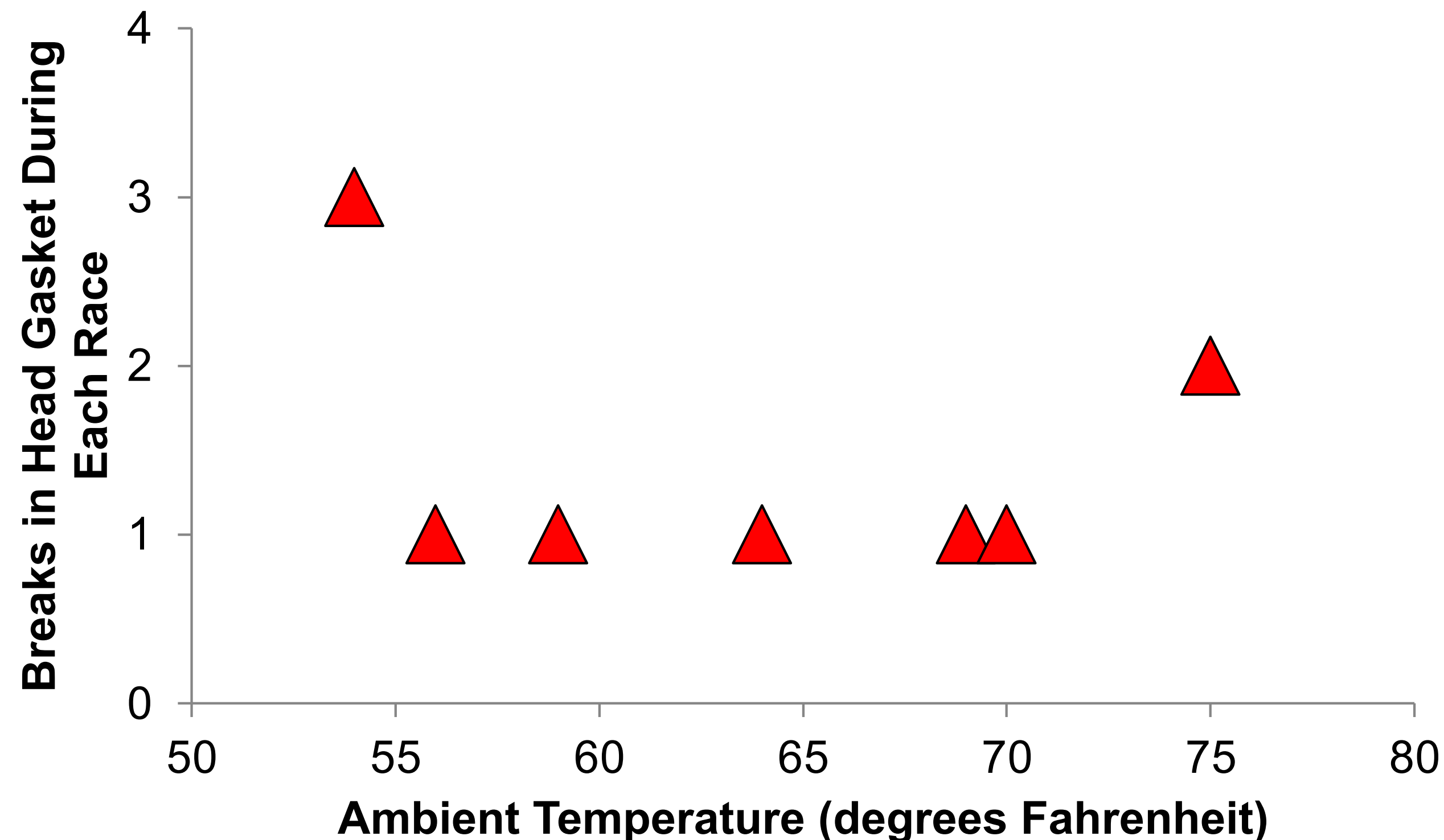
Space Shuttle Challenger

- 73 seconds into the 25th shuttle flight (Jan 28, 1986) suffered a failure of the SRB o-rings caused by low temperature at launch (26 deg F)
- 7 dead: Scobee, Smith, McNair, Resnik, Jarvis, McCauliffe; estimated \$3.2 billion loss
- NASA was operating under severe budget pressure. The decision makers involved considered the shuttle program's success vital to future congressional support.
- This was a media event to drum up more support for the program. Prior delays had pushed the Challenger launch back to the point where it would have to be scrubbed because of other scheduled launches (i.e., there were no second chances).
- Within NASA shuttle launches were widely considered routine and the vehicle safe.
- The record of 24 successful launches was a powerful decision context. Because of these previous successes, engineers were asked to prove a failure was going to occur, not that success was questionable

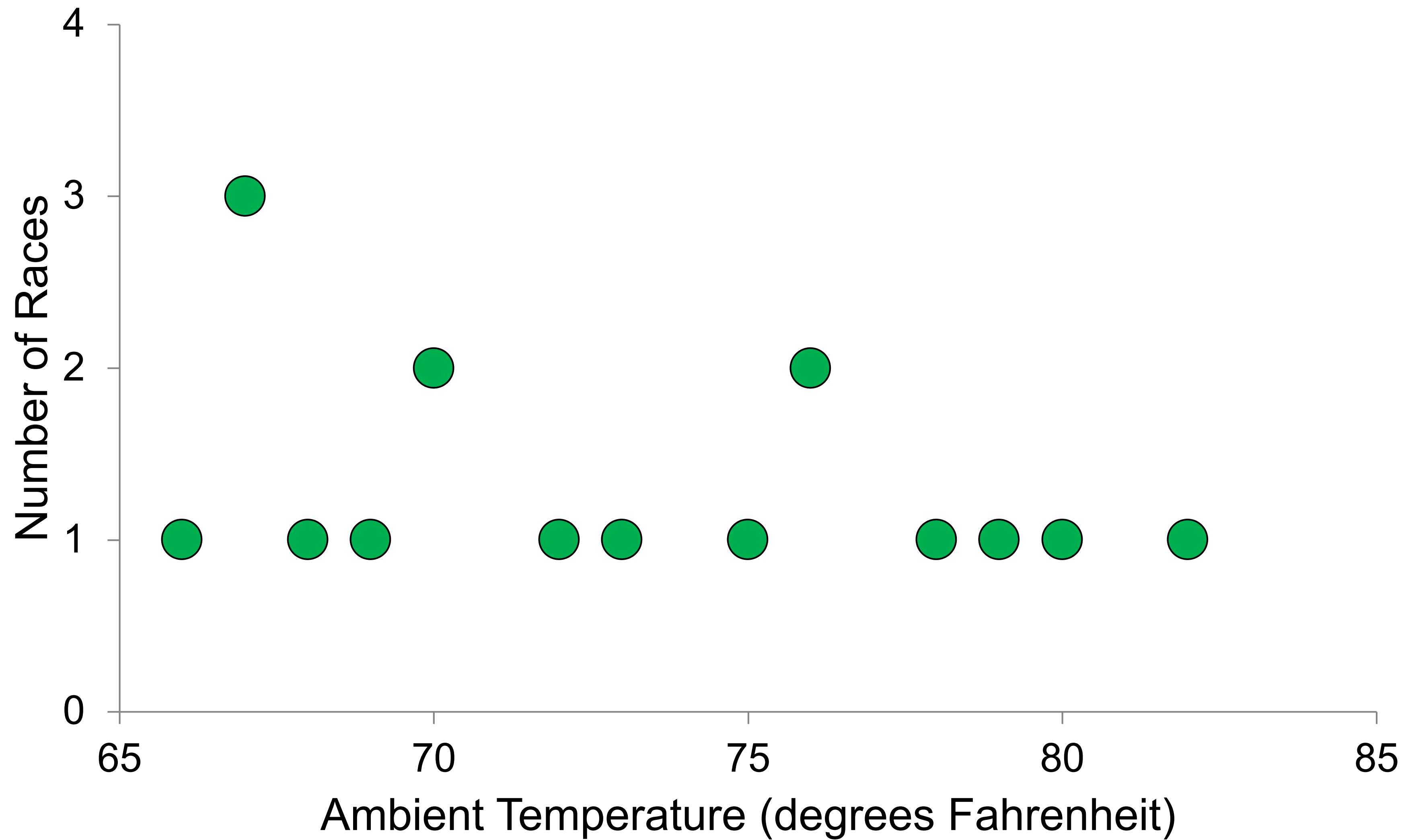


Note from Tom Davis

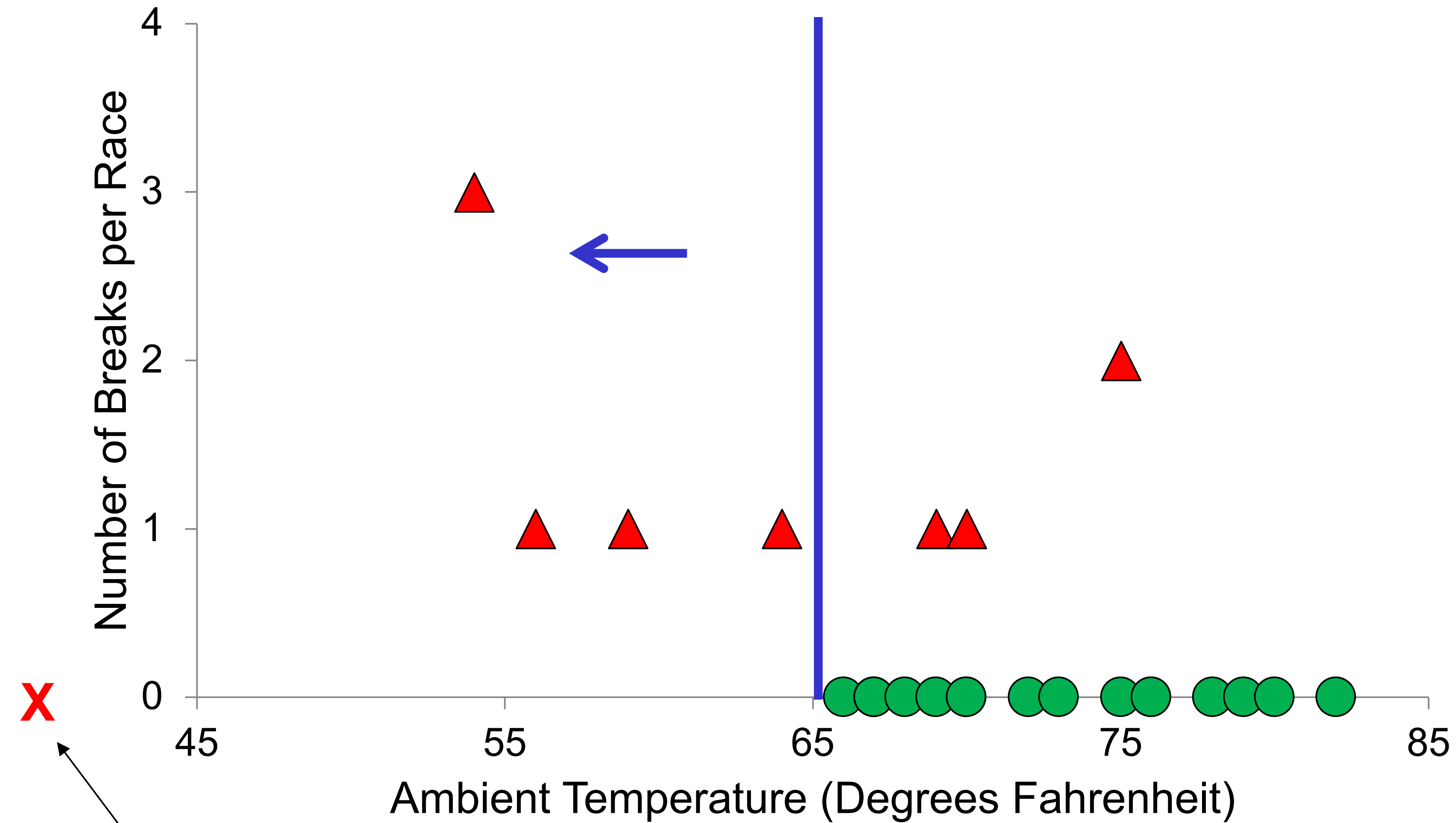
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Temperature and *no* gasket failure



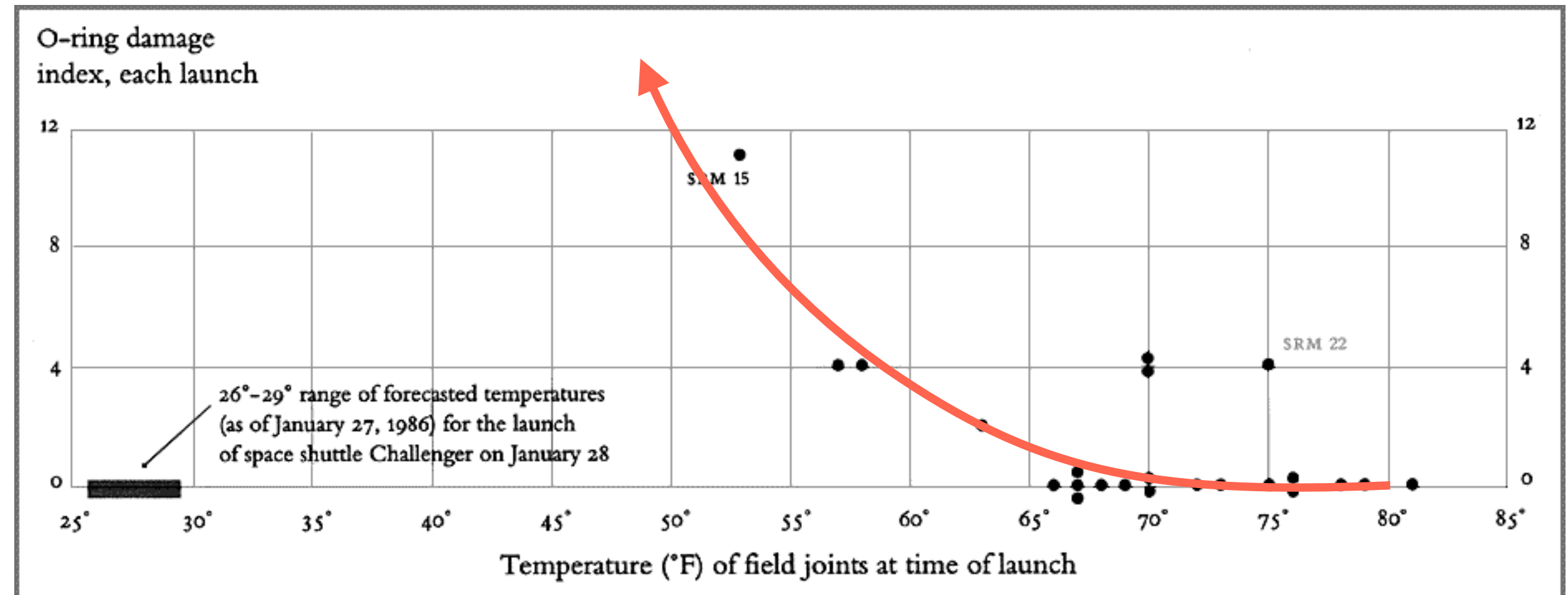
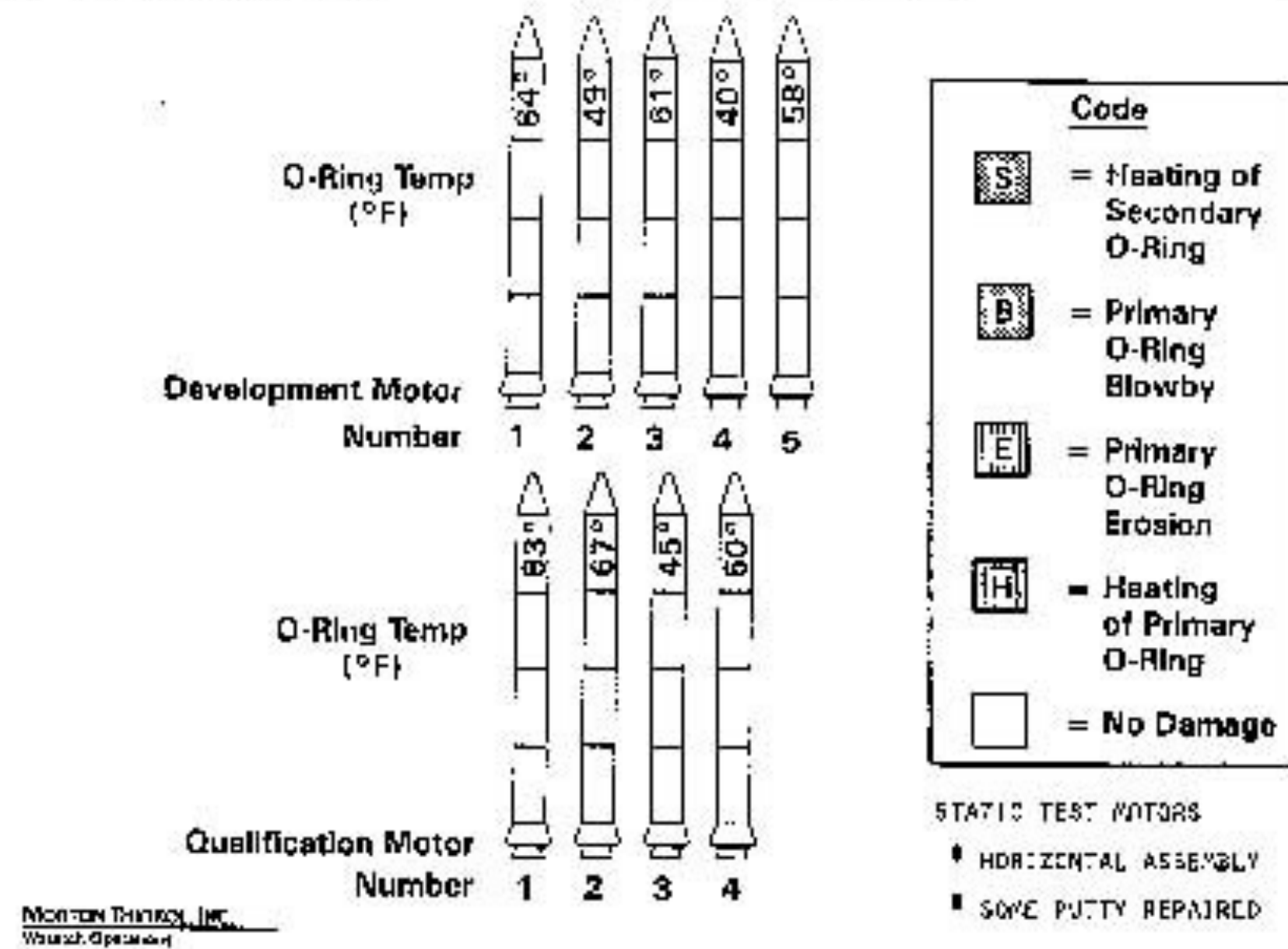
Temperature and all races



temperature that morning

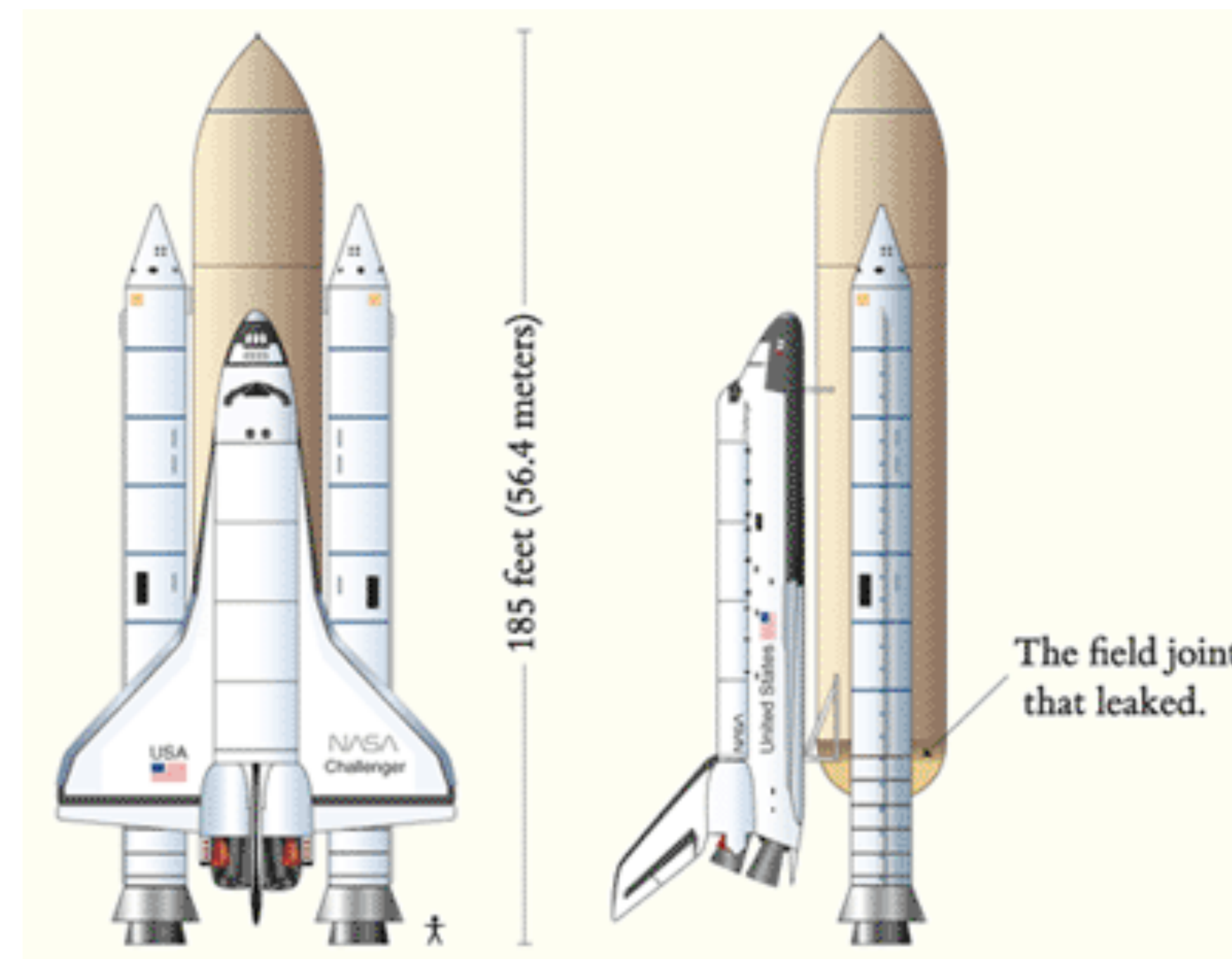
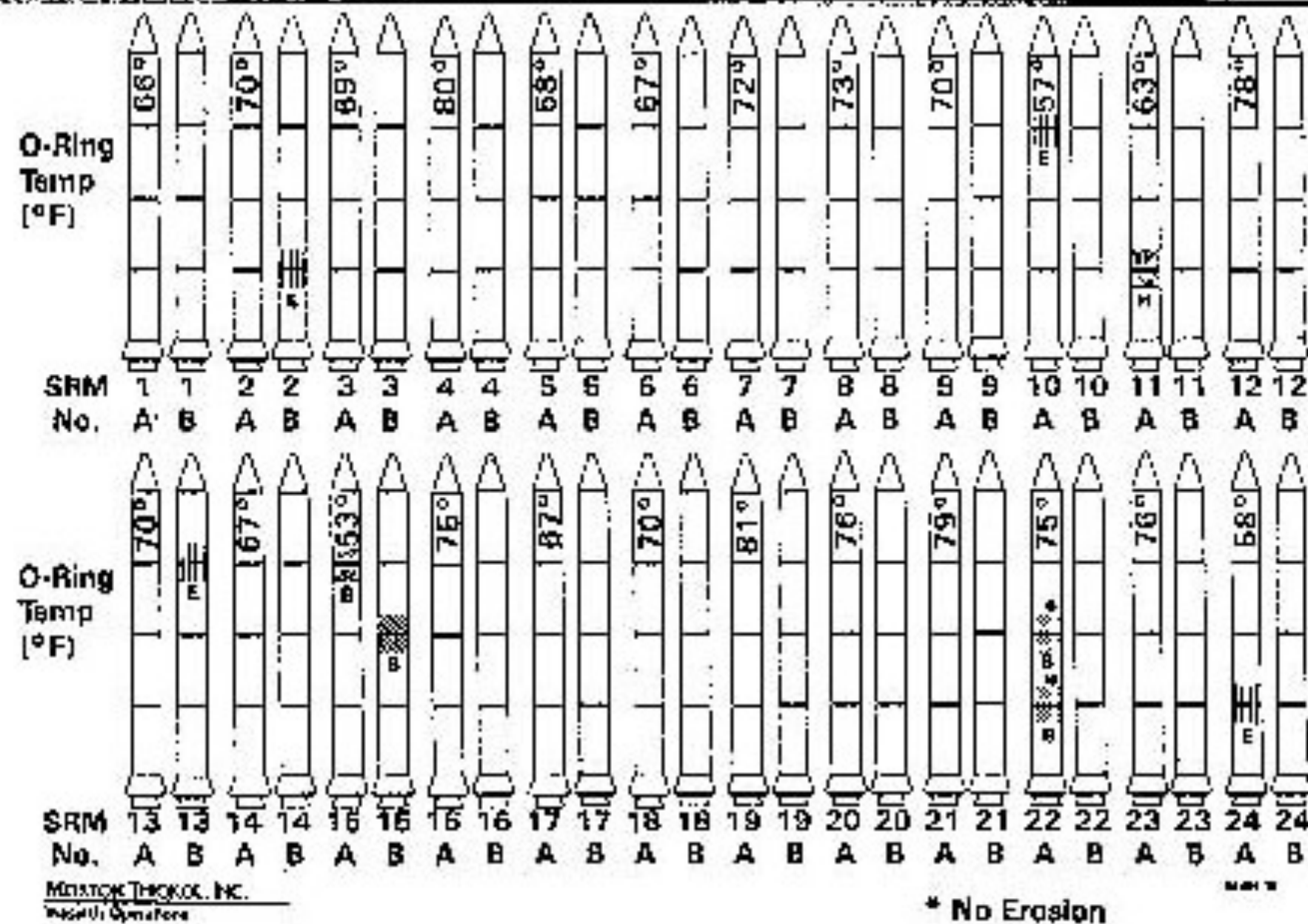
Errors of communication: Teamwork and comm failures

History of O-Ring Damage in Field Joints



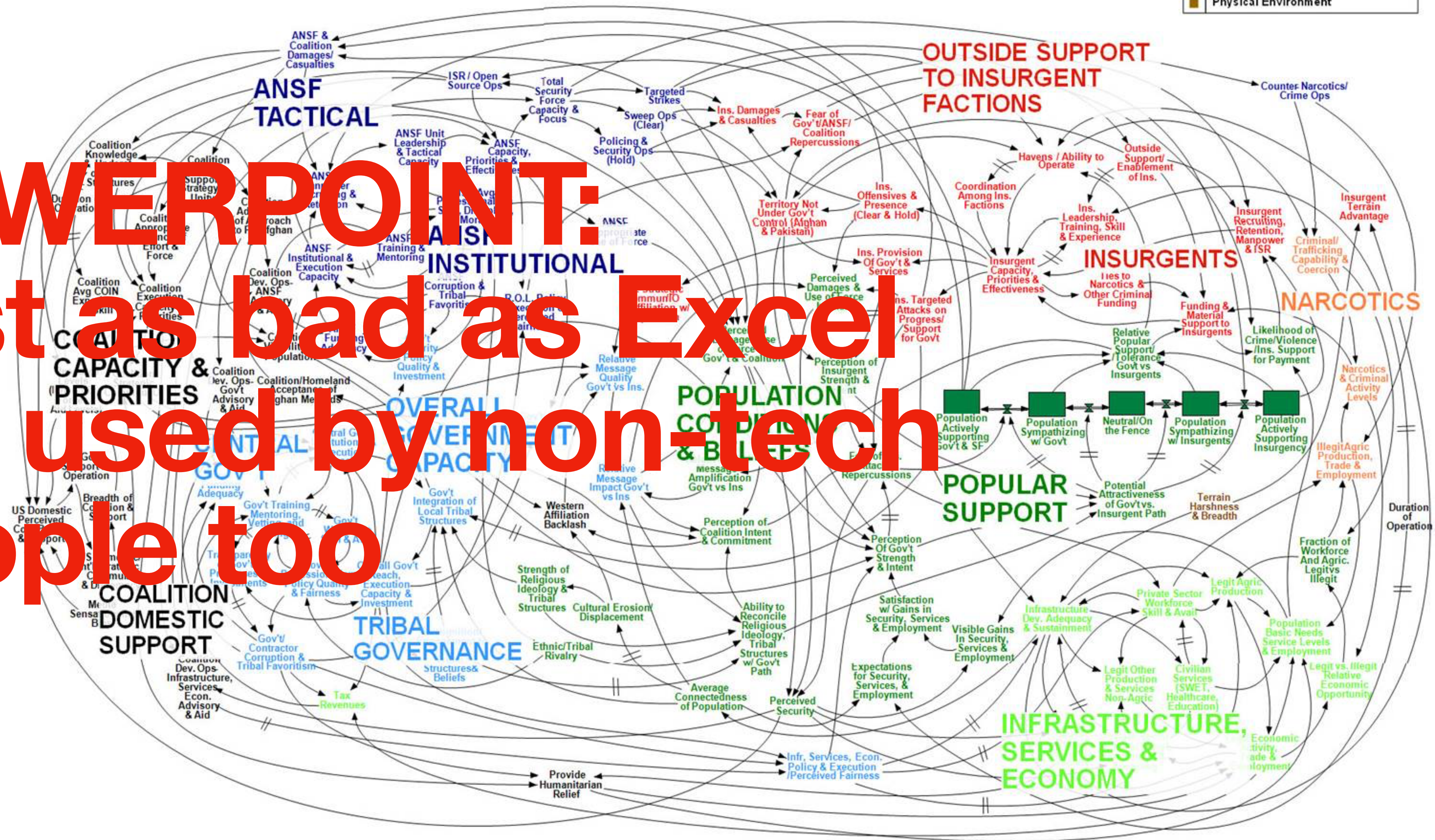
Images and graphs from Edward Tufte

History of O-Ring Damage in Field Joints (Cont)



 = Significant Delay

- POWERPOINT:
Just as bad as Excel
but used by non-tech
people too**



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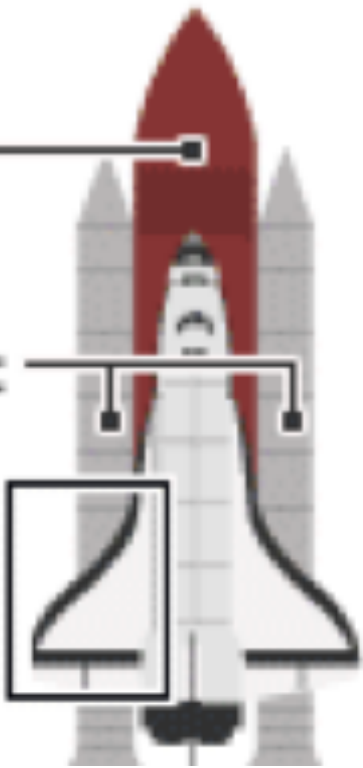
1. Launch: 16 Jan 2003

Foam from external tank strikes wing



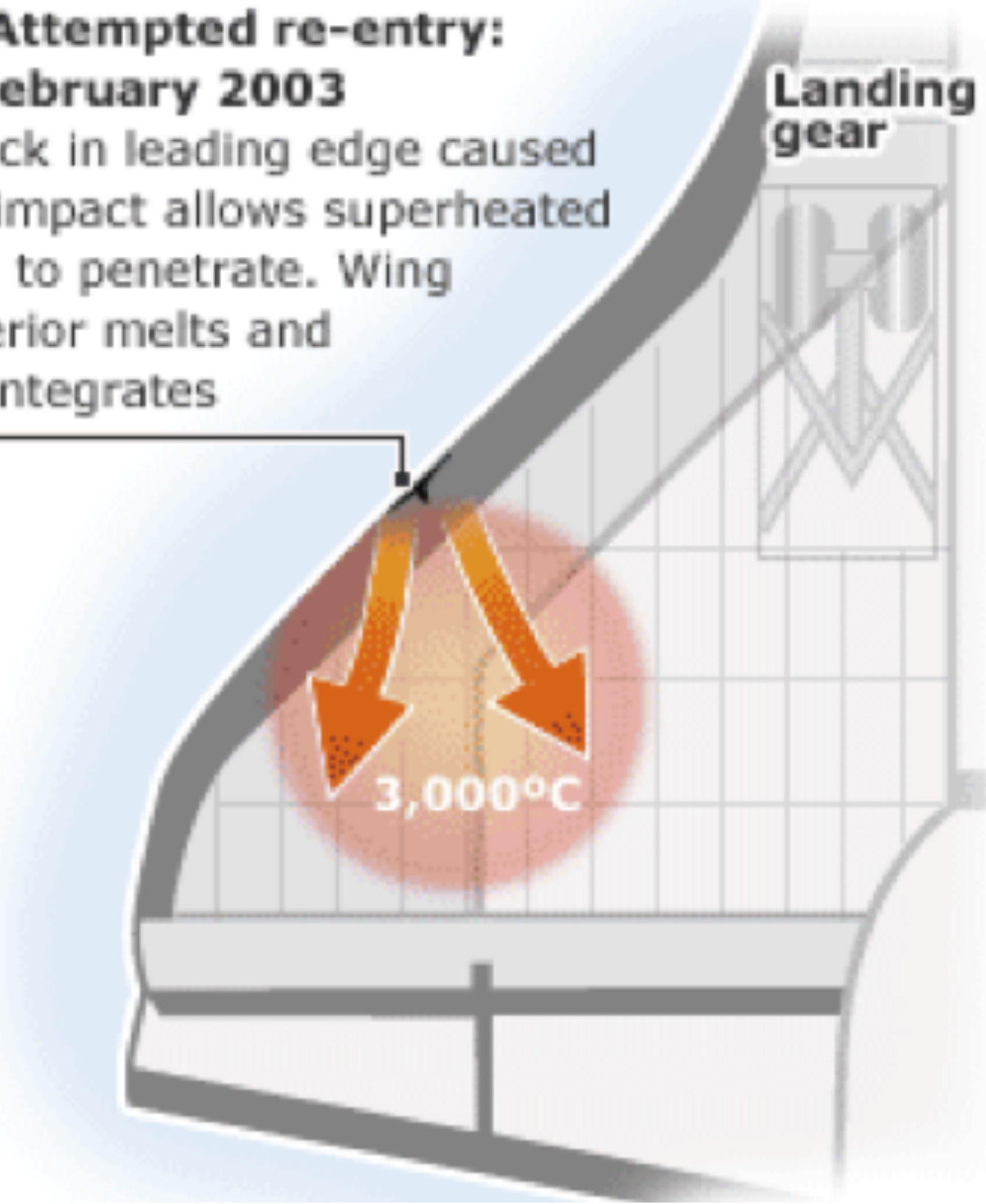
External
fuel tank

Solid rocket
boosters



2. Attempted re-entry: 1 February 2003

Crack in leading edge caused by impact allows superheated gas to penetrate. Wing interior melts and disintegrates



On this one Columbia slide, a PowerPoint festival of bureaucratic hyper-rationalism, 6 different levels of hierarchy are used to display, classify, and arrange 11 phrases:

- Level 1 Title of Slide
- Level 2 ● Very Big Bullet
- Level 3 — big dash
- Level 4 ♦ medium-small diamond
- Level 5 • tiny square bullet
- Level 6 () parentheses ending level 5

The analysis begins with the dreaded Executive Summary, with a conclusion presented as a headline: "Test Data Indicates Conservatism for Tile Penetration." This turns out to be unmerited reassurance. Executives, at least those who don't want to get fooled, had better read far beyond the title.

The "conservatism" concerns the *choice of models* used to predict damage. But why, after 112 flights, are foam-debris models being calibrated during a crisis? How can "conservatism" be inferred from a loose comparison of a spreadsheet model and some thin data? Divergent evidence means divergent evidence, not inferential security. Claims of analytic "conservatism" should be viewed with skepticism by presentation consumers. Such claims are often a rhetorical tactic that substitutes verbal fudge factors for quantitative assessments.

As the bullet points march on, the seemingly reassuring headline fades away. Lower-level bullets at the end of the slide undermine the executive summary. This third-level point notes that "Flight condition [that is, the debris hit on the Columbia] is significantly outside of test database." How far outside? The final bullet will tell us.

This fourth-level bullet concluding the slide reports that the debris hitting the Columbia is estimated to be $1920/3 = 640$ times larger than data used in the tests of the model! The correct headline should be "Review of Test Data Indicates Irrelevance of Two Models." This is a powerful conclusion, indicating that pre-launch safety standards no longer hold. The original optimistic headline has been eviscerated by the lower-level bullets.

Note how close readings can help consumers of presentations evaluate the presenter's reasoning and credibility.

The Very-Big-Bullet phrase fragment does not seem to make sense. No other VBB's appear in the rest of the slide, so this VBB is not necessary.

Spray On Foam Insulation, a fragment of which caused the hole in the wing

A model to estimate damage to the tiles protecting flat surfaces of the wing

Review of Test Data Indicates Conservatism for Tile Penetration

- The existing SOFI on tile test data used to create Crater was reviewed along with STS-87 Southwest Research data
 - Crater overpredicted penetration of tile coating significantly
 - ♦ Initial penetration to described by normal velocity
 - Varies with volume/mass of projectile (e.g., 200ft/sec for 3cu. In)
 - ♦ Significant energy is required for the softer SOFI particle to penetrate the relatively hard tile coating
 - Test results do show that it is possible at sufficient mass and velocity
 - ♦ Conversely, once tile is penetrated SOFI can cause significant damage
 - Minor variations in total energy (above penetration level) can cause significant tile damage
 - Flight condition is significantly outside of test database
 - ♦ Volume of ramp is 1920cu in vs 3 cu in for test

Here "ramp" refers to foam debris (from the bipod ramp) that hit Columbia. Instead of the cryptic "Volume of ramp," say "estimated volume of foam debris that hit the wing." Such clarifying phrases, which may help upper level executives understand what is going on, are too long to fit on low-resolution bullet outline formats. PP demands the shorthand of acronyms, phrase fragments, and clipped jargon in order to get at least some information into the tight format.

Edward Tufte

Our models are irrelevant

Debris hitting the wing was **640x** larger than the experimental data used to build these models

We have **no clue** what will happen on re-entry

Communication is key

Identify audience & setting

Identify key insight, main points of evidence, and assumptions

Organize into a story focussed on 

Create supporting visualizations

Revise to be as precise and concise as possible

Broader questions

How can we be more right and less wrong?

Being aware of our biases; external review and checks to minimize

Thinking about tools; know their strengths and limitations

Worry about process and rigorous methods more than outcomes

Social factors influence decision making; good comm and procedures are essential

Be a systems level thinker! Stay curious! Question yourself and your team as much as possible